Sensor dan Tranduser

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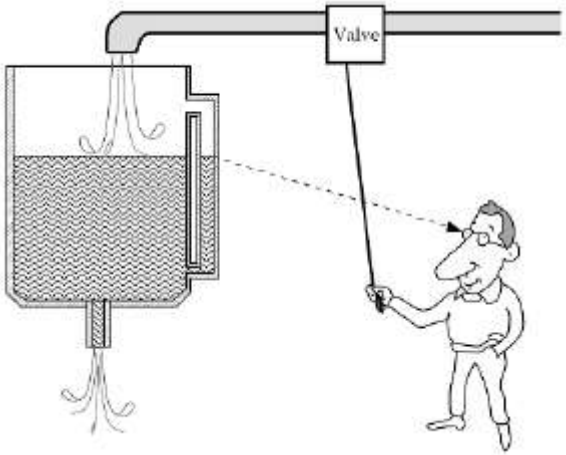
Pendahuluan

Definisi Sensor :

“Perangkat yang menerima stimulus dan merespon dengan mengeluarkan sinyal elektronik”

atau

Perangkat yang merubah besaran fisis menjadi besaran listrik.



Keluaran sesnsor :

- Tegangan
- Arus
- Tahanan

Definisi Tranduser :

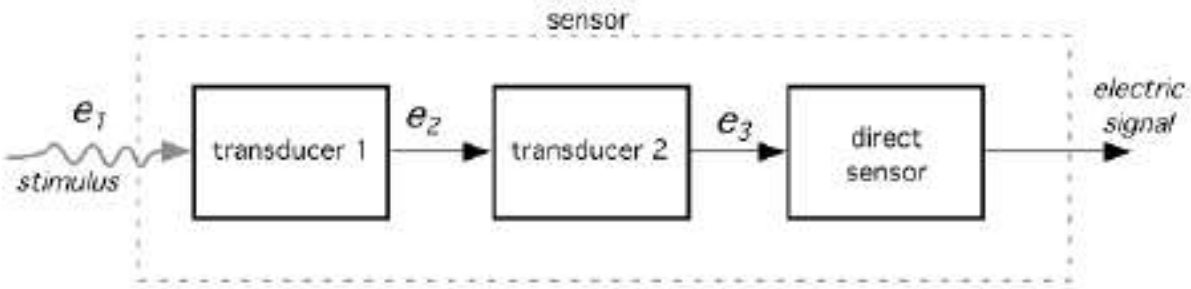
Mengubah besaran fisis menjadi besaran fisis yang lain seperti energi.

Tranduser dapat menggunakan aktuator.

Aktuator : mengubah besaran elektrik menjadi besaran fisis

Contoh aktuator : motor mengubah energi listrik menjadi mekanik

Contoh :



Sensor kimia mengubah energi reaksi kimia menjadi panas.

Thermophile mengubah panas menjadi besaran listrik (tegangan)

Jenis sensor :

1. Langsung (direct)

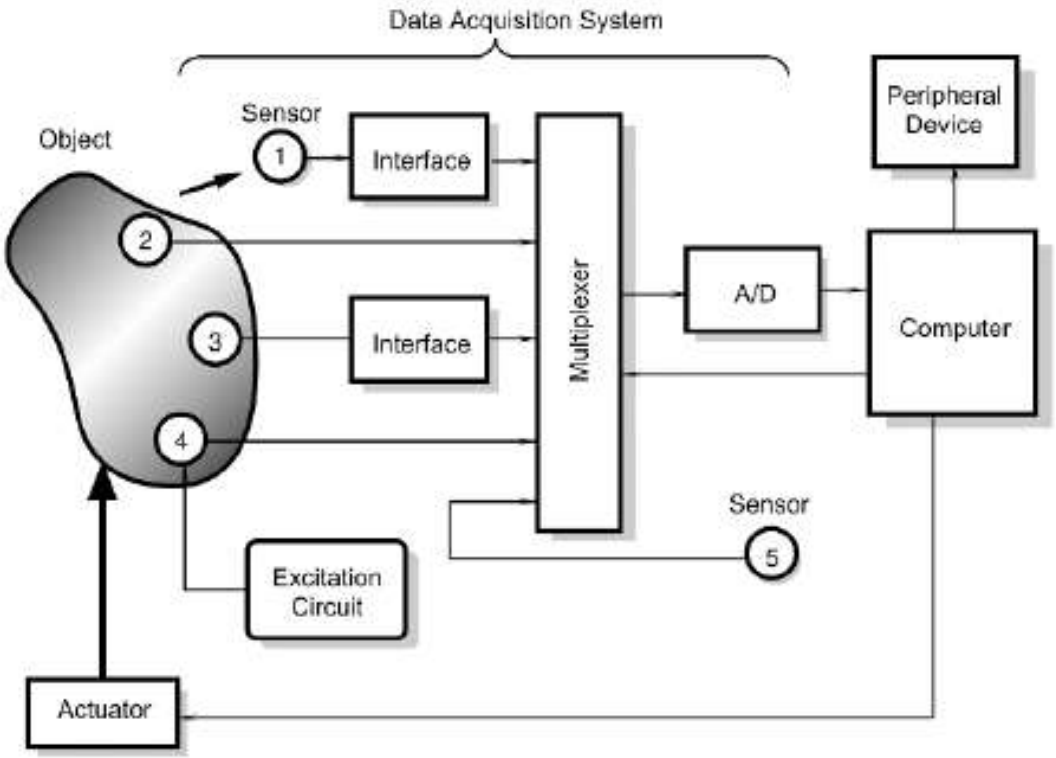
   Mengubah stimulus menjadi sinyal listrik atau modifikasi sinyal listrik oleh efek fisika

2. Sensor kompleks

Mengubah stimulus menjadi besaran listrik lewat satu atau beberapa tranduser.

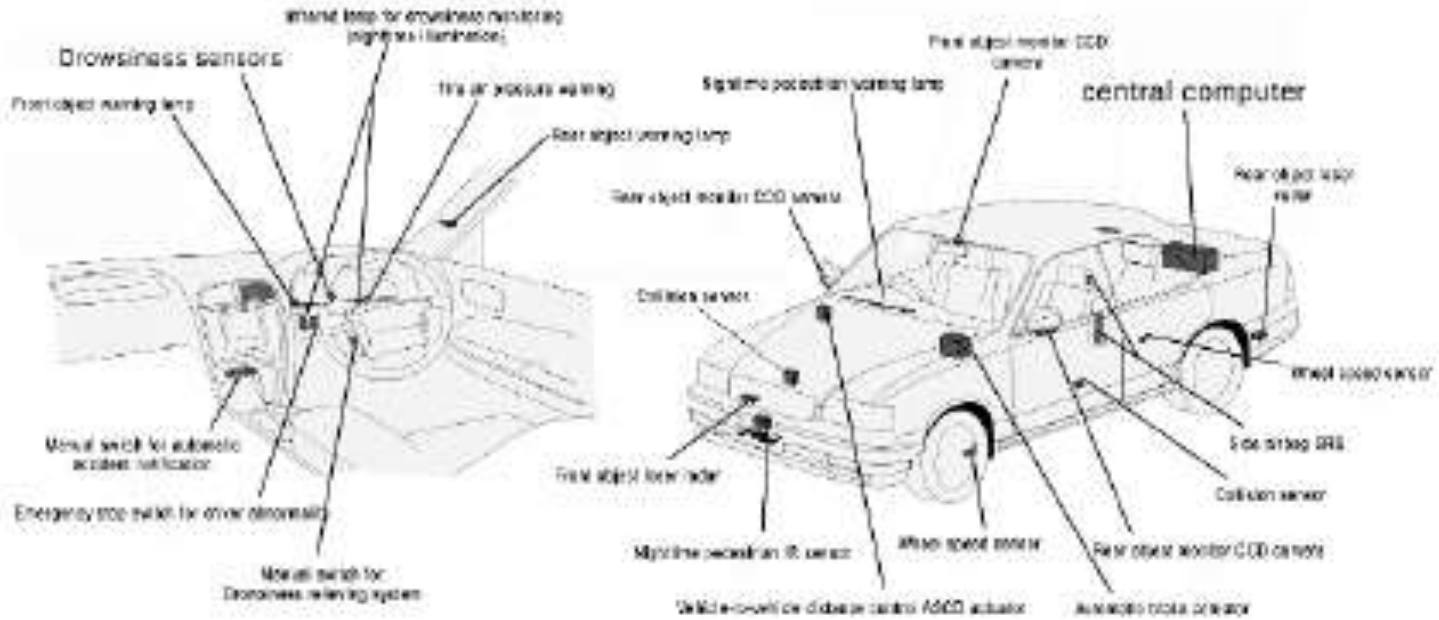
Sistem data akuisisi :

1. Sensor
2. Pengolah sinyal
3. Pemroses sinyal
4. Perangkat memori
5. Aktuator/Penampil



- Sensor 1 : Noncontact
- Sensor 2 & 3 : pasif
- Sensor 4 : aktif
- Sensor 5 : internal

- Multiple sensor



- Klasifikasi Sensor berdasar acuan (referensi) :

1. Absolut : sensor mendeteksi stimulus pada referensi ke skala fisis bebas dari kondisi pengukuran

Cth : termistor : sensor temperatur yang sensitif resistan

Resistan scr langsung berhubungan dengan skala temperatur absolut dalam celcius/Kelvin

2. Relatif : sensor menghasilkan sinyal yang berhubungan kasus khusus

Cth : Termokopel : menghasilkan tegangan listrik sebagai fungsi dari gradien temperatur

- Klasifikasi Sensor berdasar tambahan energi:

1. Aktif : sensor yang membutuhkan tambahan energi yang disebut sinyal eksitasi

Cth : termistor, : sensor temperatur yang sensitif resistan

2. Pasif : sensor yang tidak membutuhkan tambahan energi luar

Cth : Termokopel , fotodioda, piezoelektrik

Table 1.1. Specifications

Sensitivity	Stimulus range (span)
Stability (short and long term)	Resolution
Accuracy	Selectivity
Speed of response	Environmental conditions
Overload characteristics	Linearity
Hysteresis	Dead band
Operating life	Output format
Cost, size, weight	Other

Table 1.2. Sensor Material

Inorganic	Organic
Conductor	Insulator
Semiconductor	Liquid, gas, or plasma
Biological substance	Other

Table 1.3. Detection Means Used in Sensors

Biological
Chemical
Electric, magnetic, or electromagnetic wave
Heat, temperature
Mechanical displacement or wave
Radioactivity, radiation
Other

Table 1.4. Conversion Phenomena

Physical	Chemical
Thermoelectric	Chemical transformation
Photoelectric	Physical transformation
Photomagnetic	Electrochemical process
Magnetoelectric	Spectroscopy
Electromagnetic	Other
Thermoelastic	Biological
Electroelastic	Biochemical transformation
Thermomagnetic	Physical transformation
Thermooptic	Effect on test organism
Photoelastic	Spectroscopy
Other	Other

Table 1.5. Field of Applications

Agriculture	Automotive
Civil engineering, construction	Domestic, appliances
Distribution, commerce, finance	Environment, meteorology, security
Energy, power	Information, telecommunication
Health, medicine	Marine
Manufacturing	Recreation, toys
Military	Space
Scientific measurement	Other
Transportation (excluding automotive)	

Table 1.6. Stimulus

Acoustic	Mechanical
Wave amplitude, phase, polarization	Position (linear, angular)
Spectrum	Acceleration
Wave velocity	Force
Other	Stress, pressure
Biological	Strain
Biomass (types, concentration, states)	Mass, density
Other	Moment, torque
Chemical	Speed of flow,rate of mass transport
Components (identities, concentration, states)	Shape, roughness, orientation
Other	Stiffness, compliance
Electric	Viscosity
Charge, current	Crystallinity, structural integrity
Potential, voltage	Other
Electric field (amplitude, phase, polarization, spectrum)	Radiation
Conductivity	Type
Permitivity	Energy
Other	Intensity
Magnetic	Other
Magnetic field (amplitude, phase, polarization, spectrum)	Thermal
Magnetic flux	Temperature
Permeability	Flux
Other	Specific heat
Optical	Thermal conductivity
Wave amplitude, phase, polarization, spectrum	Other
Wave velocity	
Refractive index	
Emissivity	
reflectivity, absorption	
Other	

Contoh :

Accelerometer : Alat pengukur kecepatan

Stimulus:	Acceleration
Specifications:	Sensitivity in frequency shift per gram of acceleration, short- and long-term stability in Hz per unit time, etc.
Detection means:	Mechanical
Conversion phenomenon:	Elastoelectric
Material:	Inorganic insulator
Field:	Automotive, marine, space, and scientific measurement